

Chapter – 8: Measurement of Time and Motion

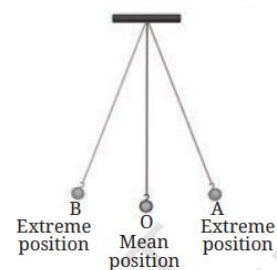
- Prerna, younger sister – watch – sport channel
- Prerna – enjoy running – fastest – district level – 100 metre sprint – interschool competition
- Now training – state level – dreams – represent India – internation level – 100 metre sprint
- Watching rerun – Olympic games – past – Prerna – amazed – measurement – time taken – so advanced
- Her school – teacher – used stopwatch – timing
- Noticed – mother – wear watch – sister – used mobile phone
- Her uncle – braille watch – recently – talking watch
- Large clock – near entrance – her thoughts – past people – no modern gadgets

Measurement of Time

- Humans – interested – keeping track – time – long ago
- Noticed – many events – nature – repeat themselves – definite interval
- Ex. – rising, setting – Sun – phases – moon – season – changing
- Used cycles – these events – timekeeping
- 1st – calendars – day – rising, setting – Sun – then – quest – ways – time of days
- Many devices – help them – measure – smaller intervals – sundials, water clocks, hourglasses, candle clocks
- Sundial – time – changing position – shadows of obj. – light of Sun
- Water clocks – flow of water – in, out – vessel – measure time
- One type – water – flows out – vessel – markings on it
- Other type – bowl – fine hole bottom – float – water surface – sink – fixed interval – lifted up – floated again
- Hourglass – time – measured – flow of sand – one bulb to another
- Candle clocks – candles – markings – indicate – passage of time – burned
- Activity –
 - Take – transparent plastic bottle – $\frac{1}{2}$ L or more – with cap
 - Cut it – 2 parts – roughly – middle
 - Use – drawing pin – small hole – bottle cap
 - Place – upper part – inverted over – lower part
 - Fill – upper part – water – may add ink – more visible level
 - Water – start dripping – lower part – use watch – mark water levels – every min.
- Water clock – ready – pour water – lower part – back into – upper part – watch water levels
- Civilisation advanced – people – travel long dist. – measurement – very critical
- Development – better mech. Devices – measurement of time – weights, gears, springs – 14th cent. onwards
- Invention – pendulum clock – 17th cent. – major breakthrough – mech. timekeeping

A simple pendulum

- Consists – small metallic ball – **bob** – suspended – long thread
- Pendulum – at rest – mean position
- Bob – moves – one side – released – oscillatory motion
- This motion – **periodic** – repeats path – fixed interval



- Pendulum – 1 oscillation – bob – mean position (O) – moves – extreme (A) – dir. change – extreme (B) – returns – mean (O)
- Pendulum – also 1 oscillation – A to B to A
- Time taken – pendulum – 1 oscillation – **time period**
- Activity –
 - Collect – string (150 cm), heavy metal ball (hooked), stopwatch, ruler
 - Tie bob – one end – string
 - Fix – other end – string – rigid support – length of string b/w – support, bob – 100 cm
 - Wait – let bob rest – pendulum – ready
 - Hold bob – move – one side – release it – take care – do not push
 - Use watch – **measure** – time – pendulum – 10 oscillations – **record** time
 - Repeat – 3-4 times
 - Divide by 10 – time taken – **calculate** time period – note down
- **Conclude** – time period – pendulum – almost same
- Time period – simple pendulum – given length – constant
- This property – used – measurement of time
- All clocks – old, modern – some process – repeat – mark – equal intervals

SI unit of time

- **SI unit** – time – **second** – **symbol (s)**
- Larger units – **minute (min)**, **hour (h)**
- 60 s = 1 min – 60 min = 1 h
- Activity –
 - Look – wall clock – smallest interval – you can measure?
- One second – smallest interval – measure – this clock

Slow or Fast

- What you mean – something – slow, fast
- Suppose – watching 100 m race – straight track
- All players – begin together – starting line – sometime later – not together
- How to **decide** – running faster?
- Someone – ahead of others – same instant – running faster
- Dist. – obj. cover – given interval – which one faster
- Faster runner – higher speed

Speed

- Compare – dist. – 2 or more obj. – unit time – which one – faster
- Unit time – 1 s, 1 min, 1 h
- Dist. covered – unit time – **speed**
- **Determine** – speed – divide – total dist. covered by time taken
 - $speed = \frac{\text{total distance covered}}{\text{total time taken}}$
- We know – SI units – length, time
- Speed – dist. / time – **SI unit** – **metre / second** – symbol – **m / s**
- Other units – **kilometre / hour** – symbol – **km / h**

- Activity –
 - Look up – railway time table – internet
 - **Identify** – train – railway station – nearest – your place
 - Find – next station – train stops – also find – dist.
 - Note – time – departure – your station – arrival – next station
 - Find diff. – calculate – time taken – train – cover dist.
 - Calculate – speed – record
 - Repeat – 4-5 – diff. train types
 - **Compare** – speeds – which – faster?
- Train – cover – max. dist. – unit time – fastest train – highest speed

Relationship between speed, distance and time

- We know – calculate speed
 - $speed = \frac{total\ distance\ covered}{total\ time\ taken}$
- Write this – diff. form – calculate dist. covered
 - $total\ distance\ covered = speed \times total\ time\ taken$
- Also calculate – time taken – cover dist.
 - $total\ time\ taken = \frac{total\ distance\ covered}{speed}$
- Above all – speed – total dist. by total time
- Obj. – might not travel – same speed – entire dur.
- Speed – we calculated – **average speed** – BUT – this book – use term ‘speed’

Uniform and Non-Uniform Linear Motion

- Remember – linear motion – chapter – measurement of length – Grade 6 – *curiosity*
- Obj. – move – straight line – motion – linear motion
- Imagine – train – track – st. line – 2 adj. railway station
- Motion of train – 2 stations – ex. – linear motion
- Train starts – A – slower speed – speeds up – slows down – D – next station
- B/w 2 stations – some dist. – B to C – const. speed
 - A _____ B _____ C _____ D
- Obj. – move – st. line – const. speed – **uniform linear motion**
- Train – uniform motion – B to C
- Obj. – move – st. line – speed – varies – **non-uniform linear motion**
- Train – non-uniform motion – A to B – C to D
- Obj. – uniform motion – equal dist. – equal intervals
- Obj. – non-uniform motion – unequal dist. – equal intervals
- Train X – equal dist. – equal intervals – uniform motion – BUT – Train Y – non-uniform motion
- Uniform motion – idealization – daily life – obj. – do not move – const. speed – long intervals – reason – average speeds